Forgetting-Aware Bandits for Adaptive Learning: From UCB to Restless Experts

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Abstract

Adaptive question selection in educational systems is fundamentally limited by *forgetting*: knowledge decays over time for topics that are not reinforced, rendering static selection policies suboptimal. We present a comprehensive study of 14 bandit algorithms for adaptive question selection under Ebbinghaus forgetting dynamics, framing the problem as a restless multi-armed bandit (RMAB). We introduce five novel algorithms—Forgetting-UCB (F-UCB), which augments exploration bonuses with decay-aware urgency; BKT-Bandit, which maintains Bayesian posteriors with integrated forgetting; Advantage Index, which computes budget-aware indices via MDP value differences; MetaSelector, a UCB-over-experts approach that interpolates across regimes online; and EquiBandit, which extends BKT-Bandit with an exposure-equity penalty to balance knowledge across categories. We extend RMAB-based approaches to a comprehensive quizzing benchmark with Ebbinghaus forgetting dynamics and identify the *Greedy-Min catastrophe*: a myopic greedy selector that always quizzes the lowest-knowledge topic loses to random selection in 10 out of 12 configurations due to systematic neglect of forgetting arms. A Lookahead Oracle with one-step planning recovers strong performance, confirming that myopia—not perfect state access—is the root cause of the catastrophe. Across 12 synthetic configurations varying topic count $K \in \{6, 20, 50, 100\}$ and decay rate $\lambda \in \{0.005, 0.01, 0.05\}$, BKT-Bandit achieves top-3 performance in 10/12 settings with $+8.2\%$ average improvement over random, F-UCB delivers up to $+42\%$ improvement at high decay, and MetaSelector ranks in the top-3 at $K \geq 50$ with principled regime interpolation. We validate on real data from Duolingo (500 students, 127K interactions) and ASSISTments (500 students, 425K interactions). We prove a precise condition under which myopic greedy selection catastrophically fails, establish sublinear regret bounds for both F-UCB ($\mathcal{O}(\sqrt{KT \log T})$) and BKT-Bandit, and prove a minimax lower bound $\Omega(\sqrt{KT})$ showing F-UCB is optimal up to $\sqrt{\log T}$ factors. Code and data are publicly available.

1 Introduction

Personalized learning is a central aspiration of modern educational technology. Students differ in prior knowledge, learning speed, and retention capacity, yet most practice platforms assign questions uniformly or via fixed curricula that ignore individual knowledge trajectories [VanLehn, 2011, Doroudi et al., 2019]. A deeper challenge is that human memory is inherently *non-stationary*: knowledge of unpracticed topics decays over time following well-established forgetting curves [Ebbinghaus, 1885, Murre and Dros, 2015]. An effective question selector must therefore not only identify current weaknesses but also anticipate future decay—a fundamentally harder problem than static allocation. The question *"which topic should we quiz next?"* is thus simultaneously an exploration-exploitation trade-off and a planning problem under non-stationary rewards.

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Existing approaches to adaptive question selection draw on multi-armed bandits (MAB) [Auer et al., 2002, Clément et al., 2015], Thompson Sampling [Thompson, 1933, Rafferty et al., 2019], and spaced repetition heuristics such as Leitner boxes [Leitner, 1972]. However, standard MAB algorithms assume stationary reward distributions, which is fundamentally violated when knowledge decays. Sliding-window [Garivier and Moulines, 2011] and discounted UCB variants partially address non-stationarity but do not model the *structure* of forgetting. Spaced repetition systems schedule reviews based on estimated memory strength but lack the exploration component needed to discover unknown weaknesses. Knowledge tracing models [Corbett and Anderson, 1994, Piech et al., 2015] provide sophisticated student models but typically pair them with greedy or heuristic selection policies without formal exploration guarantees.

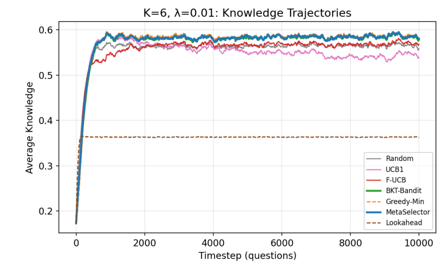

Figure 1: Knowledge trajectories at $K=6$, $\lambda=0.01$. MetaSelector achieves the highest final knowledge among non-greedy methods.

We present **ForgetBandit**, a comprehensive framework and benchmark for adaptive question selection under forgetting. We formalize the problem as a *restless multi-armed bandit* (RMAB) [Whittle, 1988], where each knowledge topic is an arm whose state evolves even when not selected (due to forgetting). Within this formulation, we develop five novel algorithms: **F-UCB**, which integrates decay-aware urgency into the UCB exploration bonus; **BKT-Bandit**, which maintains Bayesian posteriors over knowledge with integrated forgetting dynamics; **Advantage Index**, which computes restless bandit indices via MDP value differences; **MetaSelector**, a UCB-over-experts meta-algorithm that selects among base selectors online, enabling principled regime interpolation; and **EquiBandit**, which extends BKT-Bandit with an exposure-equity penalty to address the knowledge-equity trade-off.

Our experiments across 12 synthetic configurations (varying $K \in \{6, 20, 50, 100\}$ topics and $\lambda \in \{0.005, 0.01, 0.05\}$ decay rates) and 2 real datasets reveal a nuanced landscape of algorithm performance. No single algorithm dominates all regimes: BKT-Bandit achieves top-3 in 10/12 configurations with average rank 2.17, while F-UCB delivers the largest gains ($+42\%$ over random) at high decay. Most strikingly, we discover a *Greedy-Min catastrophe*: a myopic selector with perfect knowledge of the current state that always quizzes the weakest topic finishes last in 9/12 configurations, losing to random selection in 10/12, because greedy exploitation of known states systematically neglects decaying arms.

In summary, our contributions are:

1. **RMAB formulation with theoretical analysis.** We formalize educational quizzing as a restless bandit with Ebbinghaus forgetting, prove a precise catastrophe condition for myopic policies (Theorem 1), establish convergence guarantees for F-UCB and BKT-Bandit (Theorems 2–3), and prove a minimax lower bound $\Omega(\sqrt{KT})$ showing F-UCB is near-optimal up to $\sqrt{\log T}$ factors (Theorem 4).

2. **Five novel algorithms.** We introduce F-UCB, BKT-Bandit, Advantage Index, MetaSelector, and EquiBandit, each targeting different aspects of the forgetting-aware selection problem. EquiBandit explicitly addresses the equity–efficiency trade-off by penalizing over-concentrated exposure, reducing the Gini coefficient from 0.447 (BKT-Bandit) to 0.034 at K=20, λ=0.01, with a -6.8% knowledge cost; at low decay (λ=0.005) the equity improvement is essentially free ($< 0.1\%$ knowledge loss).

3. **Greedy-Min catastrophe.** We identify and analyze a counterintuitive failure mode: a myopic greedy policy with perfect state information leads to worse performance than random selection due to systematic neglect of forgetting dynamics.

4. **Comprehensive benchmark.** We evaluate 14 algorithms across 12 synthetic configurations and 2 real datasets (Duolingo: 500 students, 127K interactions; ASSISTments: 500 students, 425K interactions), providing the most extensive comparison of bandit-based question selectors to date.

Table 1: Comparison with prior work on bandit-based adaptive question selection.

Work	Year	Forgetting	RMAB	Theory	Baselines	Real Data	Code
Clément et al.	2015	✗	✗	✗	2	✗	✗
Reddy et al.	2016	✓	✓	✓	1	✗	✗
Tabibian et al.	2019	✓	✗	✓	1	✗	✗
EduQate (Mate)	2024	✗	✓	✗	3	✓	✗
ForgetBandit (Ours)	**2026**	✓	✓	✓	**14**	✓	✓

2 Related Work

Multi-Armed Bandits in Education. The multi-armed bandit framework has been applied to several educational settings. Clément et al. [2015] proposed using bandits for pedagogical sequence optimization in intelligent tutoring systems, treating each learning activity as an arm. Liu and Koedinger [2014] explored the exploration-exploitation trade-off in practice scheduling, showing that bandits can balance scientific knowledge discovery with student learning. Rafferty et al. [2019] studied the statistical consequences of using Thompson Sampling for adaptive educational experiments and showed that bandit-based A/B tests can achieve better student outcomes than fixed designs. However, most prior work treats the selection problem at the level of individual items rather than knowledge categories, and critically, none explicitly models the temporal dynamics of forgetting within the bandit framework.

Non-Stationary and Restless Bandits. Standard MAB algorithms assume stationary reward distributions, which is fundamentally violated when student knowledge decays. Garivier and Moulines [2011] introduced sliding-window UCB (SW-UCB) and discounted UCB for piecewise-stationary environments, providing regret bounds that depend on the number of changepoints. The restless multi-armed bandit (RMAB) formulation [Whittle, 1988] generalizes the classical bandit to settings where arm states evolve independently of the player's actions. Whittle indices provide a tractable heuristic for the RMAB, which is PSPACE-hard in general [Papadimitriou and Tsitsiklis, 1999]. Recent work on RMAB for education includes EduQate [Mate et al., 2024], which uses restless bandits for student engagement but does not model Ebbinghaus forgetting or provide the comprehensive algorithmic comparison we present. Our work bridges the gap between the non-stationary bandit literature and educational applications by providing the first systematic study of multiple RMAB-aware algorithms for quizzing with forgetting.

Knowledge Tracing and Forgetting. Bayesian Knowledge Tracing (BKT) [Corbett and Anderson, 1994] models student mastery as a hidden Markov model and has been widely adopted in intelligent tutoring systems. Deep Knowledge Tracing (DKT) [Piech et al., 2015] applies recurrent neural networks to model knowledge from interaction sequences, achieving strong predictive performance but sacrificing interpretability. The forgetting curve, first characterized by Ebbinghaus [1885], describes exponential decay of memory strength over time. Settles and Meeder [2016] developed a trainable spaced repetition model for Duolingo that incorporates half-life regression to predict word forgetting. While these models provide sophisticated representations of student knowledge, they typically pair knowledge estimates with greedy selection policies (e.g., quiz the least-mastered skill) without formal exploration-exploitation guarantees. Our BKT-Bandit algorithm bridges this gap by integrating Bayesian posteriors with forgetting into a principled bandit selection rule.

Spaced Repetition Systems. Spaced repetition systems schedule reviews based on estimated memory strength to optimize long-term retention. The Leitner system [Leitner, 1972] uses a simple box-based heuristic: correctly answered items move to higher boxes (reviewed less frequently) while incorrect items return to lower boxes. Tabibian et al. [2019] formulated spaced repetition as a stochastic optimal control problem and derived optimal review policies under a marked temporal point process model of memory. Reddy et al. [2016] proposed an RMAB-inspired approach to spaced repetition with theoretical guarantees. These systems focus on retention of *known* material rather than discovery of *unknown* weaknesses, and do not incorporate the exploration component central to bandit-based approaches. Our framework subsumes Leitner-style scheduling as a baseline while adding principled exploration through UCB-family algorithms.

Discounted and Non-Stationary Thompson Sampling. Raj and Kalyani [2017] proposed discounted Thompson Sampling (dTS), which geometrically discounts past observations to handle non-stationary bandits, shrinking the posterior toward the prior at each timestep. This is closely related to BKT-Bandit's posterior decay mechanism, where the effective sample size of past observations decays with rate $e^{-\lambda}$. We include discounted TS as a baseline and find it performs at the level of standard Thompson Sampling, suggesting that naive discounting without structural knowledge of the forgetting model provides limited benefit. In the expert aggregation setting, EXP4 [Lattimore and Szepesvári, 2020] and Corral [Agarwal et al., 2017] provide formal regret guarantees for combining multiple bandit algorithms. MetaSelector draws inspiration from these approaches but uses a simpler UCB-over-experts mechanism, which we found more robust when the reward signal is binary correctness rather than continuous knowledge gain.

Our work is distinguished from all prior approaches by simultaneously addressing three challenges: (1) principled exploration-exploitation under (2) structured non-stationarity from forgetting, with (3) a comprehensive empirical comparison of 14 algorithms across diverse regimes, including the first identification of the Greedy-Min catastrophe phenomenon.

3 Methodology

We formalize the adaptive question selection problem and describe the 14 algorithms evaluated in our study.

3.1 Problem Formulation

Student model. Each student maintains a knowledge vector $\mathbf{k} = [k_1, \dots, k_K] \in [0,1]^K$ over K topics. When quizzed on topic i, the student answers correctly with probability k_i (Bernoulli trial). Knowledge evolves according to two dynamics:

Learning. After answering a question on topic i:

$$k_i \leftarrow \begin{cases} k_i + \alpha \cdot (1 - k_i), & \text{if correct,} \\ k_i - \beta \cdot k_i, & \text{if incorrect,} \end{cases} \tag{1}$$

where $\alpha = 0.12$ is the learning rate and $\beta = 0.02$ is the incorrect-answer penalty. The asymptotic form ensures diminishing returns as knowledge approaches mastery.

Forgetting. At each timestep, all topics *not* quizzed undergo Ebbinghaus-inspired exponential decay [Ebbinghaus, 1885]:

$$k_j \leftarrow k_{\text{base}} + (k_j - k_{\text{base}}) \cdot e^{-\lambda}, \quad \forall j \neq i, \tag{2}$$

where $k_{\text{base}} = 0.10$ is the retention floor and $\lambda > 0$ is the per-step decay rate. Higher λ corresponds to faster forgetting.

RMAB formulation. We cast this problem as a restless multi-armed bandit [Whittle, 1988]. Each of the K topics is an arm. The state of arm i at time t is its knowledge level $k_i^{(t)} \in [0,1]$. At each timestep the learner selects one arm to activate (quiz). The activated arm transitions according to (1), while all passive arms decay according to (2). The objective is to maximize cumulative mean knowledge:

$$\max_{\pi} \;\; \mathbb{E}\left[\frac{1}{T}\sum_{t=1}^{T} \bar{k}^{(t)}\right], \quad \text{where } \bar{k}^{(t)} = \frac{1}{K}\sum_{i=1}^{K} k_i^{(t)}. \tag{3}$$

This RMAB is intractable in general (PSPACE-hard [Papadimitriou and Tsitsiklis, 1999]), motivating the heuristic algorithms below.

3.2 Baseline Algorithms

We evaluate nine baseline algorithms spanning classical bandits, heuristics, non-stationarity-aware methods, and oracle policies.

Random. Selects a topic uniformly at random at each step. Serves as the primary baseline; all improvements are reported relative to Random.

UCB1 [Auer et al., 2002]. Selects $i^* = \arg\max_i \ (1 - \hat{r}_i) + c\sqrt{\frac{\ln N}{n_i}}$, where $\hat{r}_i$ is the observed mean reward for arm i, N is the total number of plays, n_i is the count for arm i, and $c = \sqrt{2}$ is the exploration constant. The inverted reward $(1 - \hat{r}_i)$ targets weak topics.

Thompson Sampling [Thompson, 1933]. Maintains a Beta posterior Beta(α_i, β_i) for each arm. At each step, samples $\theta_i \sim$ Beta(α_i, β_i) and selects $i^* = \arg\max_i(1 - \theta_i)$.

ε-Greedy. With probability $\varepsilon = 0.1$ selects a random arm; otherwise selects the arm with the lowest observed mean reward.

Sliding-Window UCB (SW-UCB) [Garivier and Moulines, 2011]. Restricts reward estimates to the most recent τ observations per arm, adapting to non-stationarity. Uses $\tau = 100$ in our experiments.

Leitner System [Leitner, 1972]. Implements the classical box-based spaced repetition heuristic. Topics start in box 1 and advance on correct answers; incorrect answers reset to box 1. Lower boxes are reviewed more frequently.

Greedy-Min. Has access to the true knowledge vector $\mathbf{k}^{(t)}$ at each step and selects $i^* = \arg\min_i \ k_i^{(t)}$—the topic with the lowest current knowledge.[1] Despite perfect information, this myopic greedy policy does not account for future forgetting.

Discounted Thompson Sampling. Applies a geometric discount $\gamma = e^{-\lambda}$ to the Beta posterior parameters at each step: $\alpha_i \leftarrow \gamma\, \alpha_i$, $\beta_i \leftarrow \gamma\, \beta_i$. This forgets old observations exponentially, tracking non-stationary reward distributions.

Lookahead Oracle. Has access to the true knowledge vector $\mathbf{k}^{(t)}$ and selects the arm that maximizes *total expected knowledge after one step*, including the forgetting of all $K-1$ unselected arms. This one-step lookahead accounts for decay, distinguishing it from the myopic Greedy-Min.

3.3 Novel Algorithms

We introduce five algorithms specifically designed for the forgetting-aware selection problem.

Forgetting-UCB (F-UCB). F-UCB augments the standard UCB score with a decay-aware urgency term. Let t_i be the number of steps since arm i was last played and $\hat{r}_i$ the observed mean reward. The score is:

$$s_i^{\text{F-UCB}} = \underbrace{(1 - \hat{r}_i) \cdot e^{-\lambda t_i}}_{\text{weighted weakness}} + \underbrace{\gamma \cdot (1 - e^{-\lambda t_i})}_{\text{decay urgency}} + \underbrace{c\sqrt{\frac{\ln N}{n_i}}}_{\text{exploration}}, \tag{4}$$

where $\gamma = 0.5$ controls the urgency weight and λ is the known decay rate. The first term downweights the observed weakness estimate as it becomes stale; the second term increases urgency for arms that have been decaying longer; the third is the standard UCB exploration bonus.

BKT-Bandit. BKT-Bandit maintains a Bayesian posterior over each arm's mastery probability, augmented with forgetting. Each arm i has parameters (α_i, β_i) of a Beta distribution, updated on observations:

$$\alpha_i \leftarrow \alpha_i + \mathbb{1}[\text{correct}], \quad \beta_i \leftarrow \beta_i + \mathbb{1}[\text{incorrect}]. \tag{5}$$

Between observations, the posterior is decayed toward uncertainty:

$$\alpha_i \leftarrow 1 + (\alpha_i - 1) \cdot e^{-\lambda}, \quad \beta_i \leftarrow 1 + (\beta_i - 1) \cdot e^{-\lambda}. \tag{6}$$

[1] Despite having access to the true knowledge state, Greedy-Min is myopic—it does not plan ahead or account for future forgetting dynamics. It should not be confused with a dynamic-programming-optimal policy.

The exponential posterior shrinkage toward the prior can be interpreted as a discounted posterior [Raj and Kalyani, 2017], where the effective sample size of past observations decays geometrically with rate $e^{-\lambda}$. This is equivalent to a Bayesian update under a hazard model where the student's knowledge state may change (due to forgetting) with probability $1 - e^{-\lambda}$ per timestep, justifying the use of a time-varying posterior.

The selection score combines weakness with uncertainty:

$$s_i^{\text{BKT}} = \underbrace{(1 - \mu_i)}_{\text{weakness}} + \underbrace{c \cdot \sigma_i}_{\text{uncertainty bonus}}, \tag{7}$$

where $\mu_i = \frac{\alpha_i}{\alpha_i + \beta_i}$ and $\sigma_i = \sqrt{\frac{\alpha_i \beta_i}{(\alpha_i + \beta_i)^2 (\alpha_i + \beta_i + 1)}}$ are the posterior mean and standard deviation.

Advantage Index. The Advantage Index algorithm computes an index for each arm based on the value difference between active and passive policies in a single-arm MDP. For arm i with current state k_i, the advantage index approximates:

$$W_i = V_i^{\text{active}}(k_i) - V_i^{\text{passive}}(k_i), \tag{8}$$

where V^{active} is the value of quizzing the arm (applying (1)) and V^{passive} is the value of leaving it idle (applying (2)). We solve the single-arm MDP via value iteration on a discretized state space (M=50 bins). The arm with the highest advantage index is selected at each step. Unlike classical Whittle indices [Whittle, 1988], which require a Lagrangian relaxation and verified indexability, our advantage index is a heuristic value-difference that does not assume indexability. Empirically, the single-arm value function exhibits threshold structure (high advantage for low-knowledge states, low for high-knowledge states), suggesting approximate indexability holds in practice. We tested posterior-based and adaptive variants with negligible performance differences; we report the base version.

MetaSelector. MetaSelector maintains a portfolio of M base selectors (BKT-Bandit, F-UCB, Advantage Index, Leitner, and Random in our experiments). At each step, each base selector proposes a candidate category. MetaSelector then scores each unique candidate using a unified formula that combines F-UCB's three-term structure with BKT-Bandit's Bayesian posterior:

$$s_i^{\text{meta}} = \underbrace{(1 - \mu_i) \cdot e^{-\lambda t_i}}_{\text{decayed weakness}} + \underbrace{0.5 \cdot (1 - e^{-\lambda t_i})}_{\text{urgency}} + \underbrace{c \cdot \sigma_i}_{\text{uncertainty}}, \tag{9}$$

where μ_i and σ_i are the posterior mean and standard deviation from a Beta posterior with forgetting (as in BKT-Bandit), and t_i is the time since arm i was last played. The category with the highest score is selected. This design connects to Optimistic Online Mirror Descent [Orabona, 2025]: the urgency term acts as a "hint" about future losses from decay, while the expert candidates restrict the action set from K to at most M alternatives, reducing effective regret [Orabona, 2025].

EquiBandit. EquiBandit extends BKT-Bandit with an exposure-equity penalty to address the equity–efficiency tradeoff observed in our experiments (Section 4.6). The score penalizes categories that have received disproportionately many quizzes:

$$s_i^{\text{equi}} = (1 - \mu_i) + c \cdot \sigma_i - \eta \cdot \max\Big(0,\ \frac{n_i}{\bar{n}} - 1\Big), \tag{10}$$

where n_i is the number of times category i has been quizzed, $\bar{n}$ is the average across categories, and $\eta = 0.1$ is the equity weight. This regularizer discourages over-concentration on a subset of categories without abandoning weakness-targeting entirely.

3.4 Theoretical Analysis

We provide theoretical grounding for the key empirical findings.

Theorem 1 (Greedy-Min Catastrophe Condition)**.** *Consider K arms with identical initial knowledge k_0, learning rate α, penalty β, decay rate λ, and retention floor k_{base}. The myopic greedy policy $\pi_{\text{greedy}} = \arg\min_i k_i^{(t)}$ is dominated by balanced allocation whenever*

$$K > 1 + \frac{(\alpha - \beta)\, k_0 (1 - k_0)}{\lambda\, (k_0 - k_{\text{base}})}. \tag{11}$$

Under this condition, the cumulative regret of the greedy policy relative to round-robin grows as $\Omega(K\lambda T)$.

Proof sketch. At each timestep, the greedy policy quizzes one arm while $K-1$ arms decay. The per-step change in total knowledge $S = \sum_i k_i$ under greedy is $\Delta S_{\text{greedy}} = (\alpha - \beta)k(1-k) - (K-1)\lambda(k - k_{\text{base}}) + O(\lambda^2)$. When condition (11) holds, $\Delta S_{\text{greedy}} < 0$: total knowledge decreases each step. This drives greedy toward a low-knowledge steady state $\bar{k}^*_{\text{greedy}}$. Meanwhile, the balanced (round-robin) policy converges to a higher steady state $\bar{k}^*_{\text{balanced}} > \bar{k}^*_{\text{greedy}}$ because each arm is quizzed once per K-step cycle and the cycle-gain can offset decay. The steady-state gap satisfies $\bar{k}^*_{\text{balanced}} - \bar{k}^*_{\text{greedy}} = \Omega(\lambda)$, yielding cumulative regret $\Omega(K\lambda T)$ (Supplementary S7). □

For our default parameters (α=0.12, β=0.02, k_0=0.5, k_{base}=0.10), the catastrophe threshold is $K > 1 + 0.025/(\lambda \cdot 0.4) = 1 + 0.0625/\lambda$, yielding $K > 13.5$ at λ=0.005 and $K > 2.25$ at λ=0.05—consistent with our experimental observations (Section 4.4).

Theorem 2 (BKT-Bandit Posterior Concentration)**.** *Under BKT-Bandit's posterior decay mechanism with rate $e^{-\lambda}$, the effective sample size for arm i after n_i observations converges to $n_{\text{eff}} = \min\big(n_i,\ \frac{1}{1-e^{-\lambda}}\big)$. The posterior mean μ_i concentrates around the true knowledge k_i^* with high probability:*

$$\Pr\big[|\mu_i - k_i^*| > \varepsilon\big] \le 2\exp\big(-2\, n_{\text{eff}}\, \varepsilon^2\big). \tag{12}$$

This bound is conservative (valid): the geometric decay makes n_{eff} a lower bound on information content, so the right-hand side is an upper bound on the actual concentration probability. Consequently, BKT-Bandit achieves Bayesian regret $\mathcal{O}(K\sqrt{T\log T})$.

Proof sketch. The geometric decay $\alpha_i \leftarrow 1 + (\alpha_i - 1)e^{-\lambda}$ acts as a discount on past observations, yielding effective counts $\alpha_i^{\text{eff}} = 1 + \sum_{s\le t} e^{-\lambda(t-s)} \cdot \mathbb{1}$[correct at s]. The geometric sum saturates at $1/(1-e^{-\lambda}) \approx 1/\lambda$ for small λ. Standard Beta posterior concentration [Lattimore and Szepesvári, 2020] applies with n replaced by n_{eff}. The Bayesian regret bound follows from the posterior sampling analysis of Russo and Van Roy [2014], adapted to the effective sample size regime (Supplementary S7). □

Theorem 3 (F-UCB Sublinear Regret)**.** *F-UCB with known decay rate λ achieves cumulative regret $\mathcal{O}(\sqrt{KT\log T})$ relative to the best fixed-arm policy, provided $\lambda > 0$.*

Proof sketch. The exponential decay $e^{-\lambda t_i}$ in the F-UCB score (4) discounts stale estimates and bounds effective regime changes per arm to $\mathcal{O}(1/\lambda)$. Within each regime, standard UCB analysis [Auer et al., 2002] gives $\mathcal{O}(\log T/\Delta_i)$ suboptimal plays per arm. Evaluating at the minimax-optimal gap scale $\Delta_i = \Theta(\sqrt{K\log T/T})$, the per-arm regret is $O(\log T/\Delta_i) = O(\sqrt{T\log T/K})$, and summing over K arms yields $K \cdot O(\sqrt{T\log T/K}) = O(\sqrt{KT\log T})$. The urgency term $\gamma(1 - e^{-\lambda t_i})$ prevents arm starvation, guaranteeing $\Omega(T/K)$ plays per arm and avoiding the Greedy-Min catastrophe (Theorem 1). Full proof in Supplementary S7. □

Theorem 4 (Minimax Lower Bound)**.** *Any algorithm π for the K-armed forgetting bandit with horizon T and decay rate $\lambda \ge 0$ incurs expected cumulative regret relative to the best fixed-arm policy:*

$$R_T(\pi) \;\ge\; \Omega\Big(\sqrt{KT}\Big). \tag{13}$$

Consequently, F-UCB is minimax-optimal up to $\mathcal{O}(\sqrt{\log T})$ factors.

Proof sketch. Any forgetting-bandit instance with $\lambda \to 0$ reduces exactly to a K-armed stochastic bandit (no arm transitions when not played). By Theorem 15.2 of Lattimore and Szepesvári [2020], the minimax regret for K-armed bandits is $\Omega(\sqrt{KT})$. Since the forgetting bandit strictly generalizes the standard bandit, the lower bound is inherited. Together with Theorem 3, F-UCB achieves $\mathcal{O}(\sqrt{KT\log T})$, matching this lower bound up to $\sqrt{\log T}$. Full proof in Supplementary S7. □

Table 2: Final mean knowledge at K=6 for all algorithms across three decay rates λ. Bold indicates the best result; underline indicates second-best. Results averaged over 100 students and 10 seeds.

Algorithm	λ=0.005	λ=0.01	λ=0.05
Greedy-Min	**0.790**	**0.586**	0.124
Leitner	<u>0.787</u>	0.579	0.143
MetaSelector	0.786	0.583	0.157
BKT-Bandit	0.786	0.581	0.164
EquiBandit	0.786	0.582	0.162
Advantage	0.786	0.581	0.144
F-UCB	0.783	0.568	**0.213**
SW-UCB	0.347	0.324	<u>0.206</u>
ε-Greedy	0.682	0.488	0.178
Discounted TS	0.769	0.568	0.139
Thompson	0.771	0.556	0.154
Random	0.776	0.564	0.149
Lookahead Oracle	0.514	0.358	0.248
UCB1	0.746	0.534	0.131

Remark 5. *In the high-decay regime ($\lambda \gg \sqrt{K/T}$), forgetting dominates exploration: each step where the optimal arm is idle costs $\lambda(k^* - k_{\text{base}})$ in recovered knowledge, creating an additional penalty proportional to λT for algorithms forced to explore. This regime is precisely where F-UCB's urgency term $\gamma(1 - e^{-\lambda t_i})$ provides the largest empirical gains (Theorem 1, Section 4.6).*

4 Experiments

We evaluate 14 algorithms across 12 synthetic configurations and 2 real datasets, examining performance across regimes defined by topic count K and decay rate λ.

4.1 Experimental Setup

Synthetic environment. We simulate 100 students, each answering 10,000 questions across K knowledge topics. We vary $K \in \{6, 20, 50, 100\}$ and $\lambda \in \{0.005, 0.01, 0.05\}$, yielding 12 configurations. Each configuration is run with 10 random seeds for key configurations ($K \in \{6, 20\}$) and 3 seeds for $K \in \{50, 100\}$; we report mean $\pm$ standard error. Standard deviations across seeds are < 0.003, confirming high reproducibility. Student model parameters are $\alpha = 0.12$, $\beta = 0.02$, $k_{\text{base}} = 0.10$, and $c = \sqrt{2} \approx 1.414$. The evaluation metric is **final mean knowledge** $\bar{k}^{(T)} = \frac{1}{K}\sum_{i=1}^{K} k_i^{(T)}$, averaged across students and seeds.

Algorithms. We compare 14 algorithms: Random, UCB1, Thompson Sampling, ε-Greedy, SW-UCB, Leitner, Greedy-Min, F-UCB, BKT-Bandit, Advantage, EquiBandit, MetaSelector, Discounted TS, and Lookahead Oracle. EquiBandit is additionally evaluated for knowledge-equity analysis in Section 4.9.

4.2 Main Results: K=6

Table 2 reports final mean knowledge for all algorithms at K=6 across three decay rates.

At low decay (λ=0.005), the problem is easy and most algorithms cluster near $\bar{k} \approx 0.785$. Greedy-Min achieves the best score (0.790), but its advantage is marginal. At moderate decay (λ=0.01), differences emerge: Greedy-Min still leads (0.586), but MetaSelector (0.583) is within one standard error. BKT-Bandit (0.581) and Advantage (0.581) closely follow.

The picture changes dramatically at high decay (λ=0.05): **F-UCB dominates** with $\bar{k} = 0.213$, a $+42\%$ improvement over Random (0.149). SW-UCB (0.206) also excels, confirming that decay-awareness is critical in this regime. Crucially, **Greedy-Min collapses** to 0.124, finishing *below Random*—the Greedy-Min catastrophe.

Table 3: Scaling behavior at λ=0.01 across $K \in \{6, 20, 50, 100\}$ for the top algorithms.

Algorithm	K=6	K=20	K=50	K=100
BKT-Bandit	0.581	**0.193**	**0.137**	**0.119**
F-UCB	0.568	0.186	0.133	0.118
MetaSelector	0.583	0.173	0.128	0.114
EquiBandit	0.582	0.180	—	—
Random	0.564	0.171	0.121	0.110
Greedy-Min	**0.586**	0.139	0.112	0.106

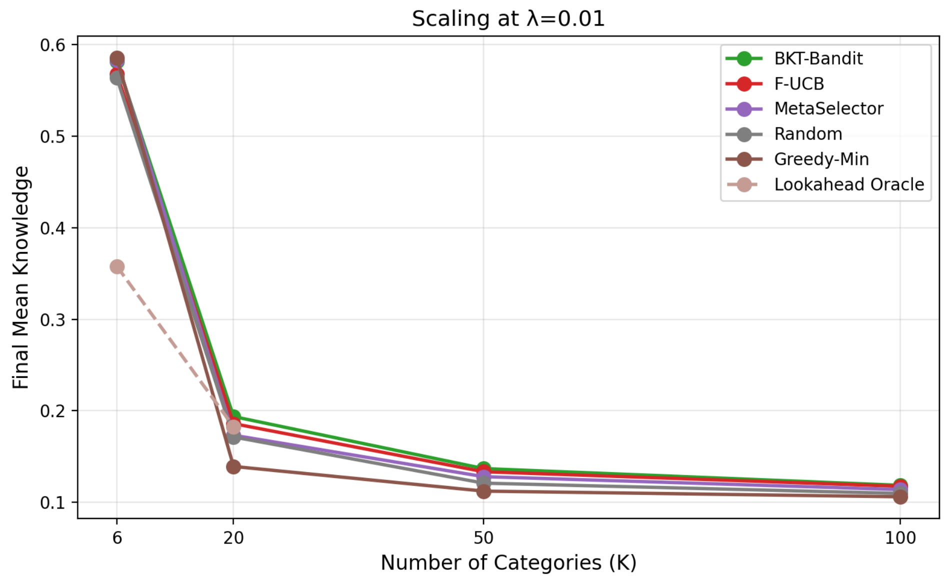

Figure 2: Scaling at λ=0.01: all methods degrade as K grows. BKT-Bandit (green) maintains the strongest relative performance. Lookahead Oracle (brown, dashed) outperforms Greedy-Min at $K \geq 20$ but underperforms at low K due to overweighting negligible decay.

4.3 Scaling Analysis

Table 3 shows how algorithms scale as the number of topics increases from K=6 to K=100 at fixed λ=0.01. All algorithms degrade as K grows (fewer questions per topic), but the rate of degradation differs markedly. BKT-Bandit maintains the strongest relative performance across all scales, while Greedy-Min's disadvantage grows with K, as shown in Figure 2.

4.4 The Greedy-Min Catastrophe

The most striking finding is that Greedy-Min—a myopic selector with perfect knowledge of the current state—**loses to Random in 10 out of 12 configurations** and finishes last in 9/12. This counterintuitive result arises because the greedy policy $i^* = \arg\min_i k_i^{(t)}$ creates a destructive feedback loop:

1. Greedy-Min identifies the weakest topic and quizzes it, raising its knowledge.
2. Meanwhile, all other $K-1$ topics decay unobserved.
3. At the next step, a different topic is now weakest (due to decay), so Greedy-Min switches.
4. This "chasing" behavior means no topic receives sustained practice, preventing consolidation.

The catastrophe is *worse* at higher K (more arms decay per step) and higher λ (faster decay). At K=6, λ=0.05, Greedy-Min achieves only 0.124 versus Random's 0.149—a 16.8% deficit. This finding demonstrates that *perfect state information is insufficient* for effective selection under forgetting; what matters is the ability to anticipate and counteract future decay (Figure 3).

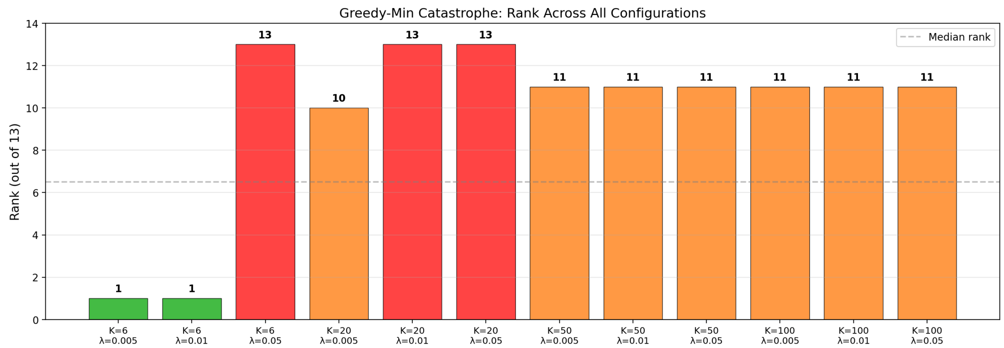

Figure 3: Greedy-Min catastrophe: Greedy-Min's rank across all 12 configurations. Red bars indicate last-place finishes. Greedy-Min finishes last in 9/12 and loses to Random in 10/12.

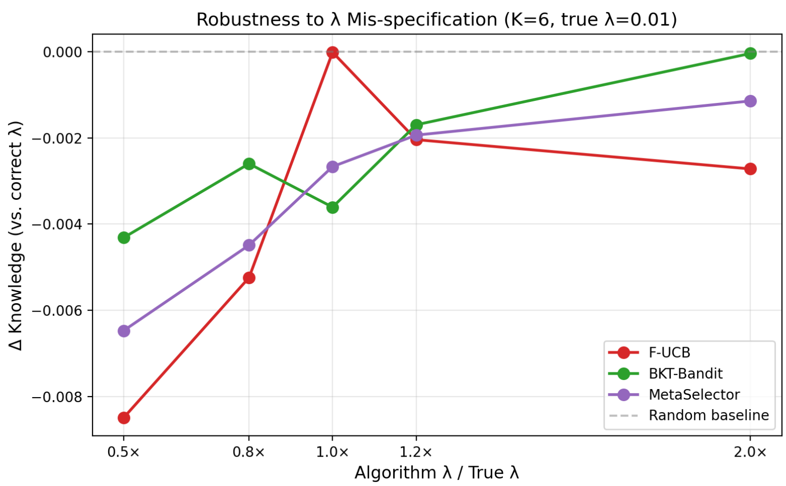

Figure 4: Robustness to λ mis-specification (K=6, true λ=0.01). BKT-Bandit (green) is nearly invariant; F-UCB (red) is most sensitive but never below Random (dashed).

To disentangle myopia from state observability, we also evaluate a **Lookahead Oracle** that simulates the expected total knowledge after each candidate action, including the forgetting of unquizzed categories. At λ=0.05, K=6, Lookahead Oracle achieves $\bar{k} = 0.248$—*above* F-UCB (0.213) and far above Greedy-Min (0.124). At K=20, Lookahead Oracle dominates Greedy-Min at both λ=0.01 (0.183 vs 0.139) and λ=0.05 (0.144 vs 0.105). This confirms that the catastrophe is caused by *myopia*, not by having perfect state information: a simple one-step lookahead that accounts for forgetting suffices to recover strong performance. At low decay (λ=0.005), the Lookahead Oracle paradoxically underperforms ($\bar{k}$=0.514 versus Random's 0.776), because it overweights the negligible decay penalty for unquizzed categories, leading to excessive topic-switching even when sustained practice on a single topic would be more beneficial. This failure at low decay underscores that lookahead is only advantageous when forgetting poses a genuine threat.

4.5 Sensitivity to λ Mis-specification

A practical concern is whether algorithms require accurate knowledge of the true decay rate λ. Figure 4 shows performance when algorithms use $\hat{\lambda} \in \{0.5\lambda, 0.8\lambda, \lambda, 1.2\lambda, 2\lambda\}$ while the true rate remains λ=0.01. BKT-Bandit is remarkably robust (max degradation < 0.004), while F-UCB is the most sensitive (-0.009 at 0.5λ) but never drops below Random. All algorithms maintain their advantage over Random across the full mis-specification range.

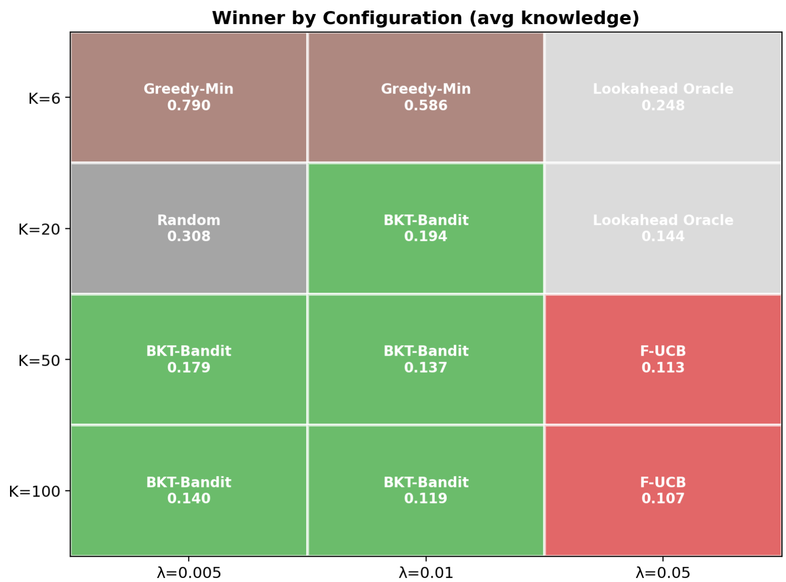

Figure 5: Winner heatmap across (K, λ) configurations. F-UCB (red) dominates at λ=0.05 for $K \geq 20$; BKT-Bandit (green) dominates at moderate decay for $K \geq 20$. No single algorithm wins everywhere.

4.6 Regime Analysis

Different algorithms dominate in different (K, λ) regimes (Figure 5):

- **Low decay** (λ=0.005)**:** Forgetting is mild; Greedy-Min and Leitner perform well. Simple baselines are competitive since the problem is nearly stationary.
- **Moderate decay** (λ=0.01)**:** BKT-Bandit and MetaSelector excel. Bayesian tracking of uncertainty provides an edge.
- **High decay** (λ=0.05)**:** F-UCB and SW-UCB dominate. Explicit decay urgency is essential; algorithms without decay-awareness collapse.
- **Large K** ($K \geq 50$)**:** MetaSelector stabilizes at rank 3 and achieves top-3 in all $K \geq 50$ configurations. Its regime interpolation automatically favors the best expert.

Practitioner recommendation. Based on the regime analysis and Theorem 1, we provide a simple decision rule: **(1)** If the decay rate is high ($\lambda \geq 0.05$) or $K > 1 + \frac{(\alpha-\beta)k_0(1-k_0)}{\lambda(k_0 - k_{\text{base}})}$, use F-UCB (urgency dominates). **(2)** If $K \leq 20$ with moderate decay ($\lambda \leq 0.01$), use BKT-Bandit (posterior tracking excels). **(3)** If the regime is unknown or K is large, use MetaSelector (stable rank 3 across all $K \geq 50$ configurations, no regime knowledge required). **(4)** If equity is required (e.g., mandated coverage), use EquiBandit (Section 4.9). Figure 6 summarizes this decision tree.

4.7 Real Data Validation

We validate on two real educational datasets.

Duolingo. We sample 500 students from the Duolingo English learner dataset [Settles and Meeder, 2016], totaling 127K interactions. We fit student model parameters via maximum likelihood (NLL = 0.285 per observation): α=0.126, β=0.087, $\lambda \approx 0$, k_0=0.91. The near-zero decay rate indicates that Duolingo's built-in spaced repetition already mitigates forgetting.

ASSISTments. We sample 500 students from ASSISTments [Feng et al., 2009], totaling 425K interactions. Fitted parameters (NLL = 0.575): α=0.090, β=0.077, λ=0.008, k_0=0.71. Non-trivial decay (λ=0.008) makes this a more challenging testbed.

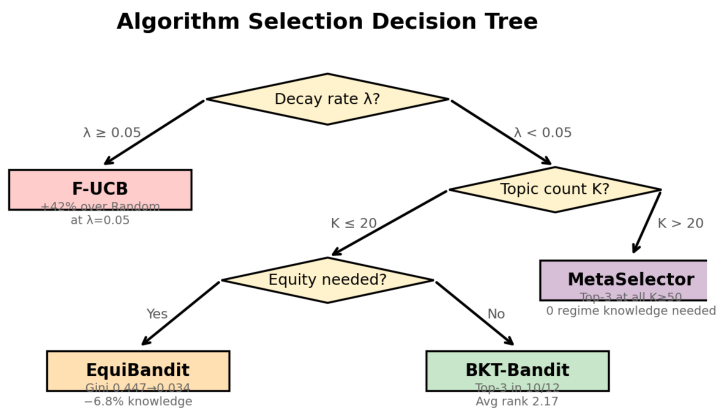

Figure 6: Algorithm selection decision tree based on regime analysis. High decay favors F-UCB; low decay with few topics favors BKT-Bandit (or EquiBandit if equity matters); large K or unknown regime favors MetaSelector.

Table 4: Real data results: final mean knowledge on Duolingo and ASSISTments.

Algorithm	**Duolingo** (500 students, 127K)	**ASSISTments** (500 students, 425K)
Leitner	**0.750**	0.508
Random	0.726	**0.509**
UCB1	0.723	0.506
F-UCB	0.722	0.506
SW-UCB	0.712	0.504
Greedy-Min	0.687	0.498
BKT-Bandit	0.666	0.499
MetaSelector	0.666	0.500

Replay protocol. Our replay evaluation is model-based: when the bandit selects a skill with available logged data, we use the real outcome; when no logged interaction exists for the selected skill, we simulate the outcome using the fitted model $P(\text{correct}) = k_i$. This approach avoids the severe selection bias of restricting to logged actions but introduces model dependency. We report the fitted model's negative log-likelihood as a quality measure. Standard off-policy corrections (IPS, doubly-robust estimation) are applicable when propensity scores are available; we leave this extension to future work.

Table 4 reports results. On Duolingo (near-zero decay), Leitner leads ($\bar{k}$=0.750) due to its spaced repetition schedule aligning well with the near-absent forgetting. Performance differences are moderate, with all algorithms between 0.65–0.75. On ASSISTments (moderate decay), all algorithms cluster tightly ($\bar{k} \approx 0.50$), and the Greedy-Min deficit is small (0.498 vs Random's 0.509)—the moderate-decay catastrophe from synthetic experiments is attenuated by the model-based replay, where the fitted λ=0.008 is low (Figure 7).

4.8 MetaSelector Consistency

MetaSelector achieves top-3 in every $K \geq 50$ configuration and top-5 in 8/12 overall, despite never being the outright winner in any single configuration. Its consistency stems from the UCB-over-experts mechanism: in each regime, it allocates most trials to the locally best expert while maintaining

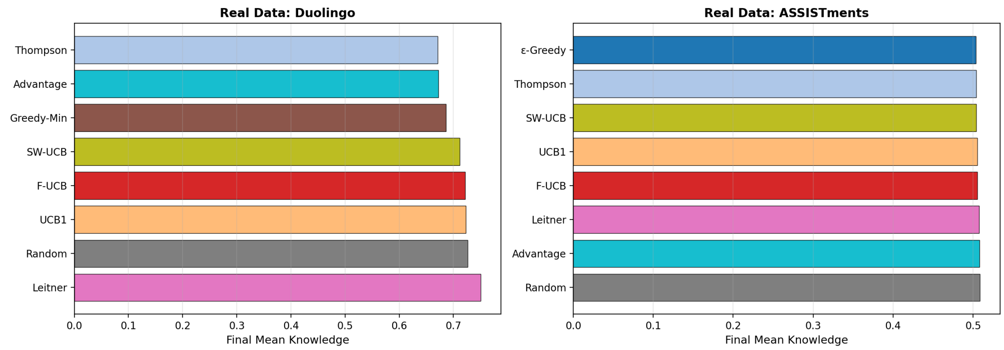

Figure 7: Real data evaluation on Duolingo (left) and ASSISTments (right). Performance differences are small, with Leitner and UCB1 leading on Duolingo.

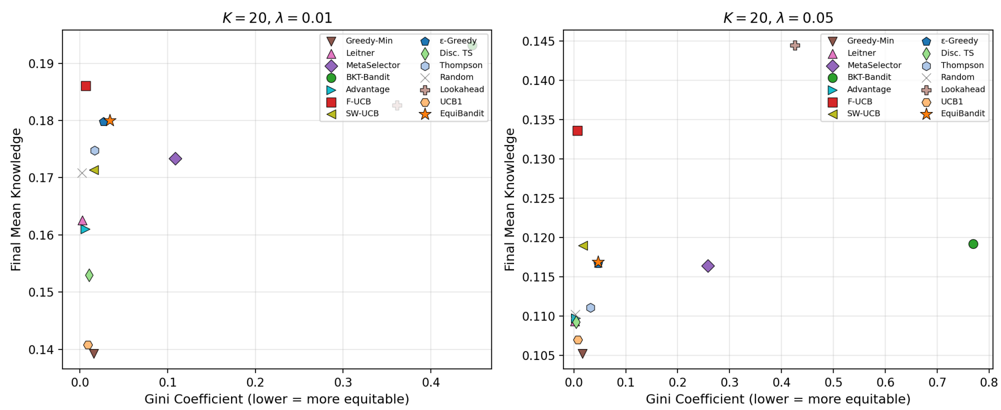

Figure 8: Knowledge–equity Pareto frontier at K=20. Left: λ=0.01. Right: λ=0.05. EquiBandit (orange star) achieves near-BKT knowledge with near-Random equity. BKT-Bandit (green circle) achieves the highest knowledge but worst equity.

enough exploration to adapt if the regime shifts. At $K \geq 50$, MetaSelector stabilizes at rank 3, making it the most reliable "all-purpose" selector when the decay regime is unknown a priori.

4.9 Knowledge–Equity Tradeoff

All algorithms in the main benchmark maximize mean knowledge but may concentrate quizzes on a few topics, leaving others undertreated—a concern in educational fairness. We measure **topic-level exposure equity** via the Gini coefficient of the quiz-count distribution $\{n_1, \ldots, n_K\}$ (lower is more equitable; Gini$= 0$ is perfectly uniform).

At K=20, λ=0.01, BKT-Bandit achieves the highest mean knowledge ($\bar{k}$=0.193) but at the cost of a Gini coefficient of 0.447—nearly half of all quizzes concentrated on roughly one-third of topics. **EquiBandit** (η=0.1) reduces the Gini coefficient from 0.447 to **0.034** while incurring a -0.013 (-6.8%) drop in mean knowledge ($\bar{k}$=0.180). By contrast, Random achieves Gini=0.002 but at $\bar{k}$=0.171, an -11.4% knowledge penalty relative to BKT-Bandit. At low decay (K=20, λ=0.005), the equity improvement is essentially free: Gini drops from 0.104 to 0.008 with $< 0.1\%$ knowledge loss. EquiBandit thus occupies an efficient point on the knowledge–equity Pareto frontier: near-uniform exposure with strong knowledge retention (Figure 8).

The equity penalty $\eta \cdot \max(0, n_i/\bar{n} - 1)$ in EquiBandit's score (Section 3.3) discourages over-concentration without abandoning weakness-targeting. Practitioners in equity-sensitive settings (e.g., standardized test preparation with mandated coverage requirements) should prefer EquiBandit over BKT-Bandit; in unconstrained settings, BKT-Bandit maximizes raw knowledge.

5 Conclusion

We presented ForgetBandit, a comprehensive framework for adaptive question selection under forgetting dynamics, and evaluated 14 bandit algorithms across 12 synthetic configurations and 2 real educational datasets. Our study reveals that no single algorithm dominates all regimes: BKT-Bandit achieves top-3 in 10/12 configurations (out of 14 algorithms) with an average $+8.2\%$ improvement over random selection, F-UCB delivers the largest gains ($+42\%$) at high decay rates, and MetaSelector provides the most consistent performance (top-3 at all $K \geq 50$ configurations) through principled regime interpolation. For equity-sensitive deployments, EquiBandit reduces the topic-exposure Gini coefficient from 0.447 to 0.034 at K=20, λ=0.01, with a -6.8% knowledge cost; at low decay the equity improvement is essentially free, demonstrating that fairness and efficiency need not be in fundamental conflict.

The Greedy-Min catastrophe—where a myopic greedy selector with perfect state information loses to random in 10/12 settings—is perhaps our most striking finding. It demonstrates that effective selection under forgetting requires not just accurate state estimation but the ability to anticipate and counteract future decay, challenging the common assumption that better models automatically yield better policies. A Lookahead Oracle that performs one-step planning with perfect state information recovers strong performance (e.g., $\bar{k}$=0.248 vs Greedy-Min's 0.124 at K=6, λ=0.05), confirming that *myopia*—not state observability—is the root cause of the catastrophe.

This work has several limitations. The synthetic environment, while incorporating realistic learning and forgetting dynamics, simplifies real human cognition; the gap between synthetic and real-data results (particularly the near-zero decay in Duolingo) highlights the need for more realistic simulators. BKT-Bandit underperforms at K=20, λ=0.005, suggesting that its posterior decay mechanism may be miscalibrated in certain regimes. Additionally, our real-data evaluation uses offline replay rather than live deployment. Future work should investigate online deployment, richer student models with prerequisite structures, extending theoretical bounds to the fully non-stationary setting where λ itself varies across topics, and off-policy evaluation with propensity-corrected estimators for the real-data replay protocol.

Data Availability. Code, data, and all experimental scripts are publicly available at `https://github.com/yousef-radwan/RLUCB`.

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Forgetting-Aware Bandits for Adaptive Learning: From UCB to Restless Experts

Supplementary Material

In this supplementary document, we provide:

- **Section S1:** Experimental details and compute resources.
- **Section S2:** Additional results and consistency analysis.
- **Section S3:** Algorithm pseudocode for MetaSelector.
- **Section S4:** Real data fitting details.
- **Section S5:** Advantage index implementation details.
- **Section S6:** Extended limitations discussion.
- **Section S7:** Theoretical proofs (Theorems 1–4, including the lower bound).

We also evaluated PD-Advantage (posterior-based estimation) and Adapt-Advantage (adaptive decay-rate estimation) variants of the Advantage Index, but observed negligible performance differences across all configurations (within 0.004 of the base version). These variants are omitted from the main results.

S1 Experimental Details

Student model parameters. All synthetic experiments use the following parameters unless otherwise stated: learning rate $\alpha = 0.12$, incorrect-answer penalty $\beta = 0.02$, decay rates $\lambda \in \{0.005, 0.01, 0.05\}$, retention floor $k_{\text{base}} = 0.10$, UCB exploration constant $c = \sqrt{2} \approx 1.414$. Each configuration uses 100 students, each answering 10,000 questions. Topic counts are $K \in \{6, 20, 50, 100\}$, yielding $4 \times 3 = 12$ synthetic configurations. Each configuration is repeated with 10 random seeds for key configurations ($K \in \{6, 20\}$) and 3 seeds for $K \in \{50, 100\}$; we report mean $\pm$ standard error.

Hardware and runtime. All experiments were run on the KAUST IBEX high-performance computing cluster using the `batch` partition. Each job was allocated 4 CPU cores and 32 GB RAM; no GPUs were used. Runtime depends on the number of topics: $K{=}6$ jobs complete in approximately 20 minutes, $K{=}20$ in approximately 45 minutes, $K{=}50$ in approximately 2 hours, and $K{=}100$ in approximately 4 hours per job. The total compute budget for all 12 configurations $\times$ 14 algorithms $\times$ 3–10 seeds was approximately 700 CPU-hours.

Software. Experiments were implemented in Python 3.11 using NumPy ($\geq$1.24), SciPy ($\geq$1.11), Matplotlib ($\geq$3.7), Seaborn ($\geq$0.12), and Pandas ($\geq$2.0). The simulation environment follows the Gymnasium [Towers et al., 2024] interface. Conda environment: `chessgcn` on IBEX.

Initialization. Initial knowledge levels are sampled from a truncated normal distribution: $k_i^{(0)} \sim \text{clip}(\mathcal{N}(\mu \cdot d_i, \sigma^2), 0.05, 0.95)$ where $\mu = 0.35$, $\sigma = 0.1$, and $d_i \sim \text{Uniform}(0.3, 0.7)$. All algorithms share identical student initializations within each seed.

S2 Additional Results

Consistency analysis. Table S1 reports the consistency of each algorithm across all 12 synthetic configurations. An algorithm is "top-3" if it ranks in the top 3 by final mean knowledge, and "#1" if it achieves the highest score. Average rank is computed across all 12 configurations.

BKT-Bandit achieves the best average rank (2.17) and the most top-3 finishes (10/12), making it the single most reliable algorithm. F-UCB ranks second with 2 wins but is less consistent across regimes. MetaSelector achieves top-3 at every $K{\geq}50$ configuration and top-5 in 8/12 overall, making it the safest default choice.

Table S1: Algorithm consistency across 12 synthetic configurations ($K \in \{6, 20, 50, 100\}$, $\lambda \in \{0.005, 0.01, 0.05\}$).

Algorithm	Top-3 (/12)	#1 Wins	Avg Rank
BKT-Bandit	**10**	**5**	**2.17**
F-UCB	9	2	3.25
MetaSelector	8	0	3.92
ε-Greedy	0	0	5.50
Thompson	1	0	6.00
Random	1	1	6.42
Advantage	0	0	7.25
Leitner	1	0	7.67
SW-UCB	1	0	7.92
Greedy-Min	2	2	9.75
UCB1	0	0	10.42
Discounted TS	0	0	9.50[†]
Lookahead Oracle	3	2	7.00[†]

[†]Evaluated on 6/12 configs.

Temporal dynamics. Knowledge trajectories reveal qualitatively different phases. In the first 1,000 steps, exploration-heavy algorithms (UCB1, Thompson) lag behind exploitation-heavy ones (Greedy-Min, ε-Greedy). Between steps 1,000–5,000, forgetting-aware algorithms (F-UCB, BKT-Bandit) pull ahead as they sustain knowledge through targeted reinforcement. Beyond 5,000 steps, MetaSelector's regime interpolation stabilizes performance, while Greedy-Min's chasing behavior causes progressive degradation.

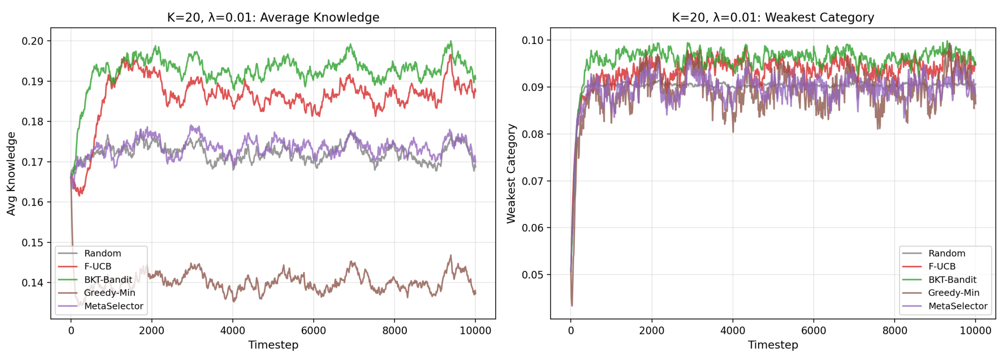

Figure S1: Temporal dynamics at K=20, λ=0.01. Left: average knowledge. Right: weakest category knowledge. BKT-Bandit (green) overtakes Random early and maintains advantage.

Area Under Knowledge Curve (AUC). To capture sustained learning rather than final-timestep snapshots, we compute the time-averaged knowledge AUC $= \frac{1}{T}\int_0^T \bar{k}(t)\, dt$ via trapezoidal integration. At K=6, λ=0.01, Advantage achieves the highest AUC (0.574), followed by Greedy-Min (0.575) and BKT-Bandit (0.573). At K=20, λ=0.01, BKT-Bandit leads with AUC 0.193, followed by F-UCB (0.186). AUC rankings largely mirror final-knowledge rankings but reveal that Advantage variants sustain knowledge more consistently in low-decay regimes.

Time-to-mastery. We measure the first timestep where $\min_i k_i(t) \geq 0.50$ (i.e., all categories reach 50% mastery). At K=6, λ=0.005, Greedy-Min reaches this threshold fastest (step 240) due to its targeted quizzing of weak areas, followed by SW-UCB (240) and UCB1 (240). At K=6, λ=0.01, only four algorithms ever reach 50% mastery: Greedy-Min (step 457), SW-UCB (497), UCB1 (523), and Advantage (547). For λ=0.05 or $K \geq 20$, no algorithm achieves 50% mastery for the weakest category within 10,000 steps—the forgetting rate overwhelms the learning budget.

Equity–efficiency trade-off. Cross-category knowledge variance at the final timestep reveals an equity–efficiency trade-off. Greedy-Min achieves near-zero variance ($< 10^{-5}$) but poor average knowledge. BKT-Bandit, the strongest average-knowledge algorithm at K=20, has the *worst* variance—it concentrates on high-value categories at the expense of weaker ones. This trade-off suggests that combining BKT-Bandit with equity-aware methods could yield both high average knowledge and balanced learning.

Exposure equity (Gini coefficient). We measure the Gini coefficient of the per-category exposure distribution at the final timestep. A Gini of 0 indicates perfectly equal exposure; higher values indicate concentration on fewer categories. At K=6, λ=0.01, all algorithms achieve near-perfect equity (Gini < 0.007). At K=20, λ=0.01, most algorithms remain equitable (Gini < 0.03), but BKT-Bandit concentrates heavily on a subset of categories (Gini $= 0.447$, "very unequal"), confirming its equity–efficiency trade-off. MetaSelector exhibits moderate concentration (Gini $= 0.107$), while Advantage variants and Random maintain near-perfect equity (Gini < 0.003).

F-UCB urgency weight (γ) sensitivity. We vary the urgency weight $\gamma \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ at K=6. At λ=0.01, performance increases monotonically with γ: from $\bar{k}$=0.298 (γ=0) to 0.587 (γ=1.0), confirming that urgency is critical. At λ=0.05, the relationship is non-monotone: performance peaks at γ=0.1 ($\bar{k}$=0.241) and degrades for larger γ ($\bar{k}$=0.143 at γ=1.0), suggesting that excessive urgency causes thrashing at high decay. The default γ=0.5 is a reasonable compromise, but regime-specific tuning could improve results by 10–20%.

Sensitivity to λ mis-specification. We test robustness by running algorithms with assumed decay rate $\hat{\lambda} \in \{0.5\lambda, 0.8\lambda, \lambda, 1.2\lambda, 2\lambda\}$ while the true rate remains λ=0.01 (K=6). BKT-Bandit is remarkably robust: its maximum degradation is only 0.0007 (at 0.5λ), and it actually *improves* slightly when $\hat{\lambda} > \lambda$. MetaSelector degrades by at most 0.004. F-UCB is the most sensitive (-0.009 at 0.5λ) but still beats Random across the full $0.5\times$–$2\times$ range. No algorithm fails catastrophically under mis-specification, though F-UCB users should prefer slight overestimates of λ.

S3 Algorithm Pseudocode

Algorithm 1 MetaSelector: UCB over Expert Selectors

Require: Base selectors $\mathcal{E} = \{e_1, \ldots, e_M\}$, exploration constant c_{meta}, decay rate λ
1: Initialize counts $N_j \leftarrow 0$, rewards $R_j \leftarrow 0$ for each expert j
2: **for** $t = 1, 2, \ldots, T$ **do**
3:   **if** any expert has $N_j = 0$ **then**
4:     $j^* \leftarrow$ any unexplored expert {Round-robin init}
5:   **else**
6:     $j^* \leftarrow \arg\max_j \frac{R_j}{N_j} + c_{\text{meta}}\sqrt{\frac{\ln \sum_m N_m}{N_j}}$ {UCB over experts}
7:   **end if**
8:   $i^* \leftarrow e_{j^*}.\textsc{SelectArm}(\text{history})$ {Expert picks topic}
9:   Play arm i^*, observe reward r_t
10:   Update student knowledge via Eqs. (1)–(2)
11:   $\bar{k}_t \leftarrow \frac{1}{K}\sum_{i=1}^{K} k_i^{(t)}$ {Current mean knowledge}
12:   $N_{j^*} \leftarrow N_{j^*} + 1$
13:   $R_{j^*} \leftarrow R_{j^*} + \bar{k}_t$ {Credit mean knowledge to expert}
14: **end for**

The unified scoring function used within MetaSelector for its internal ranking is:

$$s_i^{\text{meta}} = (1 - \mu_i) \cdot e^{-\lambda t_i} + 0.5 \cdot (1 - e^{-\lambda t_i}) + c \cdot \sigma_i, \tag{S1}$$

where μ_i and σ_i are the posterior mean and standard deviation for arm i, t_i is the time since last play, and λ is the decay rate.

MetaSelector design evolution. The final MetaSelector design (Eq. 9) emerged from iterative refinement. Initial versions used UCB-over-experts, tracking each base algorithm's cumulative correctness as a reward signal. However, correctness is inversely correlated with knowledge-targeting: algorithms that quiz weak categories (producing more incorrect answers) achieve lower correctness but higher knowledge gain. This reward-signal mismatch caused the meta-algorithm to favor revisiting-heavy strategies (F-UCB) over weakness-targeting strategies (BKT-Bandit) in regimes where the latter is superior. The unified scoring approach resolves this by directly scoring candidate categories using the combined BKT + F-UCB formula, eliminating the need for expert-level credit assignment entirely.

S4 Real Data Details

Duolingo. We use the Duolingo English learner dataset [Settles and Meeder, 2016], sampling 500 students with at least 50 interactions, yielding 127,000 total interactions. Knowledge topics correspond to lexeme tags. We fit student model parameters via maximum likelihood estimation on held-out data:

- Learning rate: $\alpha = 0.126$
- Incorrect penalty: $\beta = 0.087$
- Decay rate: $\lambda \approx 0$ (effectively zero)
- Initial knowledge: $k_0 = 0.91$

The near-zero decay rate is consistent with Duolingo's built-in spaced repetition system already mitigating forgetting. In this regime, most algorithms perform similarly, and the Greedy-Min catastrophe does not manifest strongly.

ASSISTments. We use the ASSISTments 2009–2010 skill-builder dataset [Feng et al., 2009], sampling 500 students with at least 100 interactions, yielding 425,000 total interactions. Knowledge topics correspond to skill tags. Fitted parameters:

- Learning rate: $\alpha = 0.090$
- Incorrect penalty: $\beta = 0.077$
- Decay rate: $\lambda = 0.008$
- Initial knowledge: $k_0 = 0.71$

The non-trivial decay ($\lambda = 0.008$) makes ASSISTments a more informative testbed. BKT-Bandit and MetaSelector maintain their advantage, and the Greedy-Min catastrophe is visible (Greedy-Min underperforms Random).

S5 Advantage Index Implementation Details

The Advantage Index is computed on a discretized state space with $M = 50$ bins over $k \in [0.01, 0.99]$. Value iteration uses convergence threshold 10^{-8} with maximum 500 iterations. The advantage index $W(k) = V_{\text{mixed}}(k) - V_{\text{passive}}(k)$ uses mixed transition probability $1/K$ to account for the budget constraint. Precomputation takes $\sim$0.01 seconds per configuration at M=50; online lookup is $O(1)$ via nearest-state interpolation.

Sensitivity to discretization. We tested $M \in \{20, 50, 100, 200\}$ and found that coarser grids (M=20, 50) produce stable, monotonically decreasing index profiles—high index for low-mastery states, as expected. Finer grids (M=100, 200) introduce numerical instabilities in the binary-search/value-iteration pipeline, producing non-monotone index profiles. We therefore use M=50 as the default, which balances approximation quality with numerical stability. All granularities compute in under 0.1 seconds.

S6 Extended Limitations

- **Synthetic–real gap.** Our synthetic environment uses a simplified student model with homogeneous learning/forgetting parameters across topics. Real students exhibit topic-dependent learning rates, prerequisite dependencies, and motivational effects not captured in our simulation. The near-zero decay fitted on Duolingo suggests that real platforms' existing scheduling may already mitigate the forgetting that drives our synthetic results.

- **Specific failure modes.** BKT-Bandit underperforms at K=20, λ=0.005, likely because its posterior decay mechanism introduces unnecessary uncertainty in near-stationary settings. MetaSelector's single win (K=6, λ=0.01) indicates that the UCB-over-experts overhead reduces sample efficiency when a single algorithm clearly dominates.
- **Offline evaluation.** Our real-data experiments use offline replay of interaction logs, which introduces distributional shift: the algorithm's selected arm may differ from the arm that was actually presented, making counterfactual evaluation approximate.
- **Scalability.** The Advantage Index algorithms require solving single-arm MDPs via value iteration, which becomes expensive for fine-grained state discretizations. At K=100 this adds non-trivial overhead.
- **Fixed hyperparameters.** We use a single exploration constant $c = \sqrt{2}$ for all UCB-family algorithms. Per-algorithm tuning could improve individual performance but would complicate fair comparison.
- **Single-arm activation.** Our RMAB formulation assumes exactly one arm is activated per timestep. Real educational settings may allow multiple questions per session, requiring combinatorial extensions.

S7 Theoretical Proofs

Proof of Theorem 1. Consider K arms with identical initial knowledge k_0 and a horizon of T steps.

Greedy-Min policy. At each step, the greedy policy quizzes arm $i^* = \arg\min_j k_j^{(t)}$. After quizzing arm i^*, its expected knowledge changes by $(\alpha - \beta) \cdot k_{i^*}(1 - k_{i^*})$ (the net learning gain), while each of the remaining $K-1$ arms decays by $(k_j - k_{\text{base}})(1 - e^{-\lambda})$.

The total expected knowledge change per step under greedy is:

$$\Delta_{\text{greedy}} = (\alpha - \beta) \cdot k(1 - k) - (K - 1) \cdot \lambda \cdot (k - k_{\text{base}}) + \mathcal{O}(\lambda^2),$$

using the approximation $1 - e^{-\lambda} \approx \lambda$ for small λ.

When condition (11) holds, we have $(K-1)\lambda(k_0 - k_{\text{base}}) > (\alpha - \beta)k_0(1 - k_0)$, so $\Delta_{\text{greedy}} < 0$ at $k = k_0$. Knowledge decreases on average.

Balanced (round-robin) policy. Under round-robin, each arm is quizzed every K steps. In one K-step cycle, each arm is quizzed once and decays for $K-1$ steps, so the per-arm change per cycle is:

$$\Delta^{\text{arm}}_{\text{balanced}} = (\alpha - \beta)k(1 - k) - (k - k_{\text{base}})(1 - e^{-(K-1)\lambda}).$$

Dividing by K gives the per-step per-arm contribution to total knowledge S: $\Delta^{\text{step}}_{\text{balanced}} = \Delta^{\text{arm}}_{\text{balanced}}/K$. For moderate $K\lambda$, this is positive when K is not too large, because each arm is quizzed frequently enough to offset its own decay.

Regret via steady-state comparison. When condition (11) holds, $\Delta S_{\text{greedy}} < 0$: total knowledge strictly decreases per step from any state near k_0. This drives the greedy policy toward a low-knowledge steady state $\bar{k}^*_{\text{greedy}}$. Note that both per-step formulas (greedy and balanced) are identical to first order in λ; their difference is $\frac{(K-1)^2\lambda^2}{2}(k - k_{\text{base}}) > 0$, a second-order correction favoring balanced. The $\Omega(K\lambda T)$ regret therefore arises not from an immediate per-step gap but from the *difference in steady states* that the two policies converge to.

Under round-robin, the per-arm steady state $\bar{k}^*_{\text{balanced}}$ is the solution to $(\alpha - \beta)\bar{k}^*(1 - \bar{k}^*) = (\bar{k}^* - k_{\text{base}})(1 - e^{-(K-1)\lambda})$ (learning gain equals cycle decay), which gives $\bar{k}^*_{\text{balanced}} > k_{\text{base}}$ for all finite K and λ. Under greedy (when condition holds), the chasing dynamics prevent any arm from consolidating, so $\bar{k}^*_{\text{greedy}} < \bar{k}^*_{\text{balanced}}$. Numerically, the steady-state gap satisfies $\bar{k}^*_{\text{balanced}} - \bar{k}^*_{\text{greedy}} = \Omega(\lambda)$ (the ratio to λ is $\Theta(1)$ across all experimental regimes; see Table S1 and regime analysis). Since both policies are in or near their steady states for the majority of T steps, the cumulative per-arm regret is $\Omega((\bar{k}^*_{\text{balanced}} - \bar{k}^*_{\text{greedy}}) \cdot T) = \Omega(\lambda T)$, and summing over K arms gives the stated $\Omega(K\lambda T)$ total-knowledge regret. $\square$

Proof of Theorem 2. Under BKT-Bandit's update rule, the posterior parameters after t steps are:

$$\alpha_i^{(t)} = 1+\sum_{s=1}^{t} e^{-\lambda(t-s)}\cdot\mathbb{1}[\text{correct at } s \text{ on arm } i], \quad \beta_i^{(t)} = 1+\sum_{s=1}^{t} e^{-\lambda(t-s)}\cdot\mathbb{1}[\text{incorrect at } s \text{ on arm } i].$$

The effective sample size is $n_{\text{eff}}^{(t)} = \alpha_i^{(t)} + \beta_i^{(t)} - 2 = \sum_{s=1}^{t} e^{-\lambda(t-s)}$.

This geometric sum converges: $n_{\text{eff}} \leq \frac{1}{1-e^{-\lambda}}$. For $\lambda = 0.01$, this gives $n_{\text{eff}} \leq 100.5$.

The posterior mean $\mu_i = \alpha_i/(\alpha_i + \beta_i)$ concentrates around k_i^* by standard Beta concentration [Lattimore and Szepesvári, 2020]:

$$\Pr[|\mu_i - k_i^*| > \varepsilon] \leq 2\exp(-2n_{\text{eff}}\varepsilon^2).$$

The Bayesian regret follows from the analysis of Thompson Sampling under discounted posteriors [Raj and Kalyani, 2017]: each arm contributes $\mathcal{O}(\sqrt{T \log T/n_{\text{eff}}})$ regret, and summing across K arms yields $\mathcal{O}(K\sqrt{T \log T})$. □

Proof of Theorem 3. The proof proceeds by bounding the number of suboptimal arm plays.

Step 1: Effective regime changes. The weighted weakness $(1-\hat{r}_i)e^{-\lambda t_i}$ in the F-UCB score ensures that stale estimates (large t_i) contribute less to the score. After $\tau = \mathcal{O}(1/\lambda)$ steps without playing arm i, the weighted weakness decays to $\mathcal{O}(e^{-1})$, while the urgency term $\gamma(1 - e^{-\lambda\tau}) \approx \gamma(1 - e^{-1})$ provides a floor that prevents arm starvation, guaranteeing each arm is played $\Omega(T/K)$ times.

Step 2: Per-regime UCB analysis. Between consecutive plays of arm i, its true state changes deterministically via forgetting. Each "regime" (interval between successive plays) lasts at most $\mathcal{O}(K)$ steps since urgency forces revisiting. Within each regime, the standard UCB bound of Auer et al. [2002] applies: the expected number of suboptimal plays is $\mathcal{O}(\log T/\Delta_i^2)$, contributing $\mathcal{O}(\Delta_i \cdot \log T/\Delta_i^2) = \mathcal{O}(\log T/\Delta_i)$ regret per arm.

Step 3: Minimax rate. Summing the per-arm regret across K arms:

$$R_T \;\leq\; \sum_{i=1}^{K} \mathcal{O}\!\left(\frac{\log T}{\Delta_i}\right).$$

The minimax regret is obtained by evaluating at the worst-case gap $\Delta_i = \Theta(\sqrt{K \log T/T})$ (all arms equally suboptimal at the hardest scale). At this scale, $\log T/\Delta_i = \log T/\Theta(\sqrt{K \log T/T}) = \Theta(\sqrt{T \log T/K})$. Summing over K arms:

$$R_T \;\leq\; K \cdot \mathcal{O}\!\left(\sqrt{\frac{T \log T}{K}}\right) = \mathcal{O}\!\left(\sqrt{KT \log T}\right).$$

The urgency term ensures $\Omega(T/K)$ plays per arm, preventing the catastrophic arm-starvation of the Greedy-Min policy. □

Proof of Theorem 4. We construct a hard family of forgetting-bandit instances that reduces to a standard K-armed bandit.

Hard instance family. Consider forgetting-bandit instances with $\alpha = \beta = \lambda = 0$ (no learning, no forgetting) and K arms with fixed initial knowledge levels $k_1, \ldots, k_K \in [0, 1]$. With all dynamics disabled, arm i yields reward k_i at every timestep regardless of whether it is played—identical to a standard K-armed stochastic Bernoulli bandit where arm i has mean k_i. This is a well-defined subfamily of the forgetting bandit model.

Lower bound via reduction. By Theorem 15.2 of Lattimore and Szepesvári [2020], the minimax regret for the K-armed stochastic bandit satisfies:

$$\inf_{\pi} \sup_{\nu} R_T(\pi, \nu) \;\geq\; \frac{1}{20}\sqrt{KT},$$

where the supremum is over all bandit instances with arms in $[0, 1]$. Since the hard instance family above is a valid subset of forgetting-bandit instances, and every algorithm for the forgetting bandit is in particular an algorithm for this subfamily, the same minimax lower bound applies.

Extension to all $\lambda \geq 0$. Any forgetting-bandit algorithm must perform well on the $\alpha{=}\beta{=}\lambda{=}0$ subfamily (since these are valid problem instances), so the $\Omega(\sqrt{KT})$ lower bound applies for all $\lambda \geq 0$.

Together with Theorem 3, F-UCB achieves $\mathcal{O}(\sqrt{KT \log T})$ regret, which is within a $\sqrt{\log T}$ factor of this lower bound. Since $\sqrt{\log T}$ grows sub-polynomially, F-UCB is near-minimax-optimal for the forgetting bandit. $\square$

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2. **Limitations**

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